

UNIVERSITY OF PISA

DEPARTMENT OF COMPUTER SCIENCE

Master programme in Data Science and Business Informatics

DATA MINING: FUNDAMENTAL

Project report: Seller-Level Vehicle Listings Dataset

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1 Data Understanding and Preparation

The dataset analyzed in this section comprises 13,843 records across 33 features. Each record represents an individual seller profile identified by `sp_id`, aggregating inventory and listing-level attributes associated with that seller.

In this section, we explore the data by examining feature semantics and distributions, identifying potential outliers, and outlining our strategy for handling missing values.

1.1 Data semantics

The Seller-Level Vehicle Listings Dataset contains one record per seller (`sp_id`), summarising the vehicle listings observed for that seller. The meaning, source datatype and measurement scale of each variable are reported in Table 1. The prediction target is `seller_rating_class` (low/medium/high).

Table 1: Semantic description of the dataset variables.

Feature Name	Description	Datatype	Msrmt Scale
sp_id	Unique identifier of the seller/dealer	float64	Identifier
n_listings	Total number of vehicle listings	int64	Numerical
price_median	Median advertised price of vehicles	float64	Numerical
savings_amount_median	Median advertised savings amount	float64	Numerical
daysonmarket_median	Median days vehicles remained on market	float64	Numerical
mileage_median	Median vehicle mileage	float64	Numerical
year_median	Median vehicle model year	float64	Numerical
vehicle_age_median	Median vehicle age at listing time	float64	Numerical
owner_count_median	Median number of previous vehicle owners	float64	Numerical
horsepower_median	Median vehicle horsepower	float64	Numerical
engine_displacement_median	Median vehicle engine displacement	float64	Numerical
city_fuel_economy_median	Median city fuel economy	float64	Numerical
highway_fuel_economy_median	Median highway fuel economy	float64	Numerical
back_legroom_median	Median rear-seat legroom	float64	Numerical
front_legroom_median	Median front-seat legroom	float64	Numerical
fuel_tank_volume_median	Median fuel tank capacity	float64	Numerical
height_median	Median vehicle height	float64	Numerical
length_median	Median vehicle length	float64	Numerical
width_median	Median vehicle width	float64	Numerical
pct_isCab	Proportion of vehicles previously used as cabs	float64	Numerical
pct_is_new	Proportion of vehicles classified as new	float64	Numerical
pct_salvage	Proportion of vehicles with salvage status	float64	Numerical
pct_theft_title	Proportion of vehicles with theft titles	float64	Numerical
n_listing_days	Distinct calendar days with observed listings	int64	Numerical
listing_span_days	Span in days between earliest and latest listing	int64	Numerical
dominant_body_type	Most frequent vehicle body type	str	Categorical
dominant_fuel_type	Most frequent fuel type	str	Categorical
dominant_transmission	Most frequent transmission type	str	Categorical
dominant_wheel_system	Most frequent drivetrain configuration	str	Categorical
dominant_listing_color	Most frequent exterior color	str	Categorical
franchise_status	Franchise status (franchise, independent, etc.)	str	Categorical
new_used_profile	Inventory composition (new_only, used_only, etc.)	str	Categorical
seller_rating_class	Discretized target rating (low, medium, high)	str	Ordinal

1.2 Distribution of variables and statistics

In this section, we summarise the data distributions to better understand each variable.

1.2.1 Summary of statistics

This subsection summarises the statistical properties of each attribute in the dataset. In addition to the mean, standard deviation and quartiles obtained from `df.describe()`, we report the number of missing values, skewness, kurtosis and the interquartile range (IQR) for the 24 numerical columns. Table 2 gives a high-level view of the seller-level vehicle listings data.

The table shows that the features differ considerably in scale: the `pct_*` proportions range from 0 to 1, whereas `mileage_median` spans from 0 to 238k. This disparity indicates that the data should be standardised to a common scale before being used by models that are sensitive to feature magnitude.

The table also reveals that some features are strongly skewed while others are not, and that several columns contain missing values. Both observations motivate the cleaning and preparation steps discussed in the following subsections.

Table 2: Summary statistics for the 24 numerical variables (identifier excluded).

Variable	Count	Mean	Std	Min	25%	50%	75%	Max	Missing	Skew	Kurtosis	IQR
n_listings	13843.0	118.10	137.06	5.00	25.00	70.00	168.00	4959.00	0	5.25	120.32	143.00
price_median	13843.0	23106.14	14365.07	1699.00	12495.00	22385.00	29888.00	236438.00	0	3.45	34.14	17393.00
savings_amount_median	13843.0	446.21	519.46	0.00	0.00	358.00	734.00	9333.00	0	2.40	18.30	734.00
daysonmarket_median	13843.0	49.15	56.06	0.00	26.00	36.00	51.00	1987.00	0	8.75	168.60	25.00
mileage_median	13842.0	51463.27	51845.46	0.00	25.00	33596.50	96915.75	238541.00	1	0.68	-0.77	96890.75
year_median	13843.0	2015.95	4.72	1960.00	2013.00	2017.00	2020.00	2021.00	0	-2.25	14.09	7.00
vehicle_age_median	13842.0	4.08	4.63	0.00	0.00	3.00	7.00	60.00	1	2.32	14.98	7.00
owner_count_median	13804.0	1.43	0.62	1.00	1.00	1.00	2.00	6.00	39	1.30	1.35	1.00
horsepower_median	13837.0	240.90	56.99	78.00	188.00	243.00	283.00	630.00	6	0.85	2.05	95.00
engine_displacement_median	10450.0	2925.08	808.08	1200.00	2400.00	2500.00	3500.00	6800.00	3393	1.29	2.77	1100.00
city_fuel_economy_median	13827.0	20.65	3.96	10.00	18.00	20.00	22.00	126.00	16	8.10	185.87	4.00
highway_fuel_economy_median	13827.0	27.88	3.98	15.00	25.00	27.50	30.00	105.00	16	3.88	66.79	5.00
back_legroom_median	13824.0	37.71	1.57	0.00	36.80	37.80	38.60	47.50	19	-3.53	77.81	1.80
front_legroom_median	13836.0	41.89	0.77	36.60	41.30	41.80	42.20	67.00	7	3.49	88.62	0.90
fuel_tank_volume_median	13836.0	18.35	2.93	7.15	16.40	18.25	19.20	52.00	7	1.83	7.33	2.80
height_median	13837.0	65.15	4.95	46.85	61.60	66.00	68.00	107.50	6	0.18	0.91	6.40
length_median	13837.0	191.10	10.25	151.00	184.60	189.80	192.70	282.30	6	2.18	7.59	8.10
width_median	13837.0	77.02	4.97	65.70	72.90	75.55	80.00	105.90	6	1.06	1.56	7.10
pct_isCab	13816.0	0.16	0.16	0.00	0.05	0.12	0.21	1.00	27	2.04	5.91	0.17
pct_is_new	13843.0	0.27	0.32	0.00	0.00	0.00	0.59	1.00	0	0.55	-1.38	0.59
pct_salvage	13816.0	0.01	0.06	0.00	0.00	0.00	0.00	1.00	27	10.34	119.82	0.00
pct_theft_title	13816.0	0.00	0.02	0.00	0.00	0.00	0.00	0.76	27	14.11	427.34	0.00
n_listing_days	13843.0	46.32	33.72	1.00	18.00	39.00	68.00	272.00	0	0.98	0.95	50.00
listing_span_days	13843.0	405.59	282.37	0.00	223.00	362.00	508.00	3553.00	0	2.15	9.00	285.00

1.2.2 Distribution via histograms

A histogram (Figure 1) is used to visualise the distribution of the data. This method also helps to identify patterns, anomalies and skewness, and supports the statistics reported in Table 2.

Since 24 numerical features are too many to show at this stage, we select a semantically representative subset of columns: `n_listings`, `price_median`, `daysonmarket_median`, `mileage_median`, `year_median`, `owner_count_median`, `horsepower_median`, `city_fuel_economy_median`, `front_legroom_median`, `fuel_tank_volume_median`, `height_median` and `pct_is_new`.

From the figure, most of the selected features are highly skewed, which is confirmed by the skewness values above 1 in Table 2. Most sellers have fewer than 500 listings and stay on the market for fewer than 200 days. `pct_is_new` and `mileage_median` show that many of the cars being sold are new and have never been driven, while `owner_count_median` indicates that most cars have had one or two previous owners.

The large empty space on the right side of several panels indicates extremely high (or low) outliers, which is also supported by the high kurtosis in Table 2.

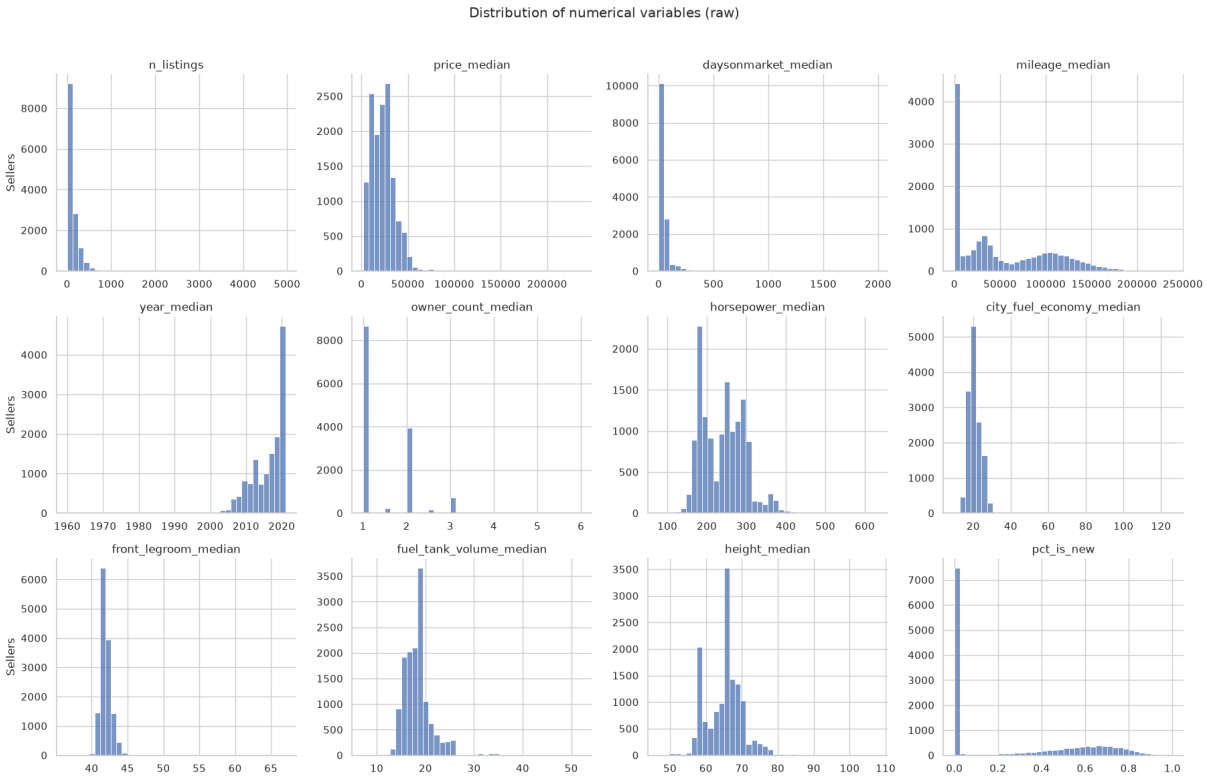


Figure 1: Histograms of the selected numerical variables.

1.2.3 Distribution of categorical features

To better understand the distribution of the categorical data, we use pie charts for low-cardinality features and bar charts for higher-cardinality features, where magnitudes are easier to compare. The pie charts are shown in Figure 2, while the bar charts are shown in Figure 3.

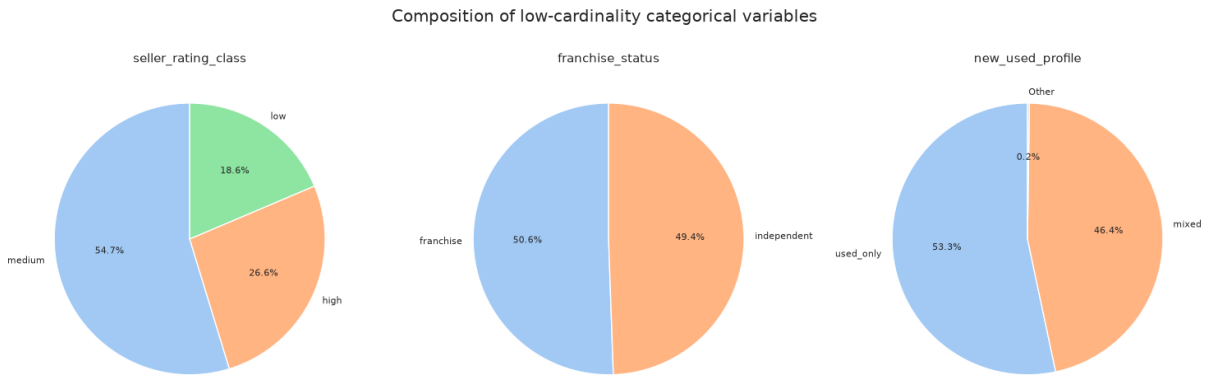


Figure 2: Composition of the low-cardinality categorical variables.

From the pie charts, `seller_rating_class` is dominated by the *medium* class, followed by *high*. In contrast, `franchise_status` and `new_used_profile` are relatively well balanced across their values.

The bar charts show that the value `a` (automatic) dominates `dominant_transmission`, while `dominant_fuel_type` is clearly dominated by *gasoline*. For `dominant_wheel_system`, we include a `(missing)` value to highlight the large proportion of missing data. In `dominant_listing_color`, black and white are the most common colours, followed by *unknown*; colours such as brown, green, gold and teal are present but represent only small fractions of the cars. Finally, SUVs account

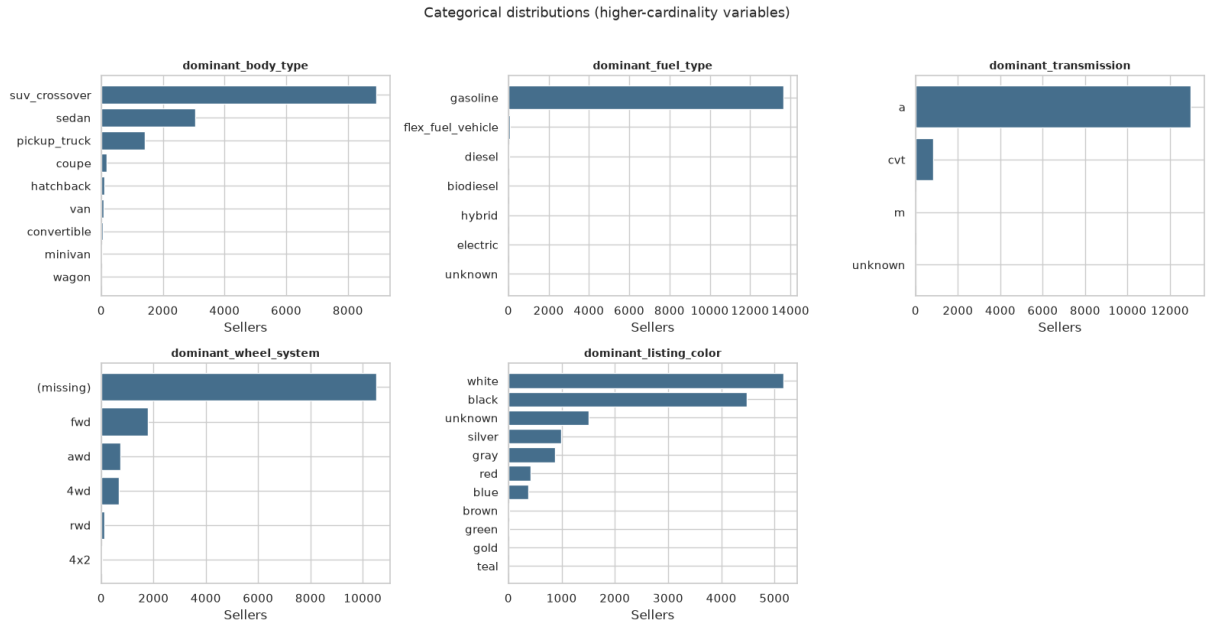


Figure 3: Distributions of the higher-cardinality categorical variables.

for a very large proportion of `dominant_body_type`, followed by sedans and pickup trucks; the remaining body types are present but not significant.

1.3 Assessment of data quality

Before statistical imputation and feature transformation, we ran a syntactic and semantic audit across the dataset. First, no duplicated rows were found, and the sellers' identifiers (`sp_id`) are unique. Second, no negative values appear in any numerical attribute. Third, all percentage attributes (`pct_isCab`, `pct_is_new`, `pct_salvage`, `pct_theft_title`) are correctly bounded within the closed interval $[0, 1]$, and advertised discounts never exceed the original prices (`savings_amount_median` $<$ `price_median`). Moreover, temporal consistency holds across all records (`n_listing_days` $\leq$ `n_listings` and `n_listing_days` $\leq$ `listing_span_days` $+1$).

We also observed that count and calendar-year medians (`year_median`, `owner_count_median`, `daysonmarket_median`) contain fractional `.5` values. These are valid and arise from sellers with an even number of vehicle listings.

The only datatype change was to `sp_id`, originally stored as `float64`, which we converted to string so that it is automatically excluded from numerical summary statistics.

1.3.1 Missing value handling

Twenty attributes contain missing values: 17 with explicit `NaN` values and 3 with implicit *unknown* values (see Table 3 and Figure 4).

We applied tailored strategies to handle the missing values in each attribute:

- `dominant_wheel_system`: dropped due to the high percentage of missing values (75.98%).
- `engine_displacement_median`: filled with the median grouped by `dominant_body_type` and `horsepower_median` quintiles, ensuring that the engine size reflects both the vehicle's body type and power.
- `dominant_listing_color`: implicit *unknown* values (10.91%) are kept as a separate category, preventing mode substitution from over-representing the *white* label.
- `dominant_fuel_type` and `dominant_transmission`: the implicit *unknown* values (1 and 6 instances) are replaced by the group mode computed across each `dominant_body_type`.
- `vehicle_age_median`: the single missing entry (0.01%) was computed as $2020 -$ `year_median`.

Table 3: Summary of explicit (`NaN`) and implicit (*unknown*) missing values, with the handling strategy.

Variable	Missing	Missing (%)	Type	Strategy
dominant_wheel_system	10,518	75.98%	NaN	Drop feature (>50% missing)
engine_displacement_median	3,393	24.51%	NaN	2D Group median (body type + hp_bin)
dominant_listing_color	1,510	10.91%	*unknown*	Retain as valid category
owner_count_median	39	0.28%	NaN	Group median (by body type)
pct_isCab, pct_salvage, pct_theft_title	27 each	0.20%	NaN	Group median (by body type)
back_legroom_median	22 (19+3*)	0.16%	NaN	Group median (*3 zeros → NaN)
city_ & highway_fuel_economy_median	16 each	0.12%	NaN	Group median (by body type)
front_legroom_, fuel_tank_volume_median	7 each	0.05%	NaN	Group median (by body type)
height_, length_, width_, horsepower_	6 each	0.04%	NaN	Group median (by body type)
dominant_transmission	6	0.04%	*unknown*	Group mode (by body type)
dominant_fuel_type	1	0.01%	*unknown*	Group mode (by body type)
vehicle_age_median	1	0.01%	NaN	Mathematical (2020 − year)
mileage_median	1	0.01%	NaN	Group median (by body type)

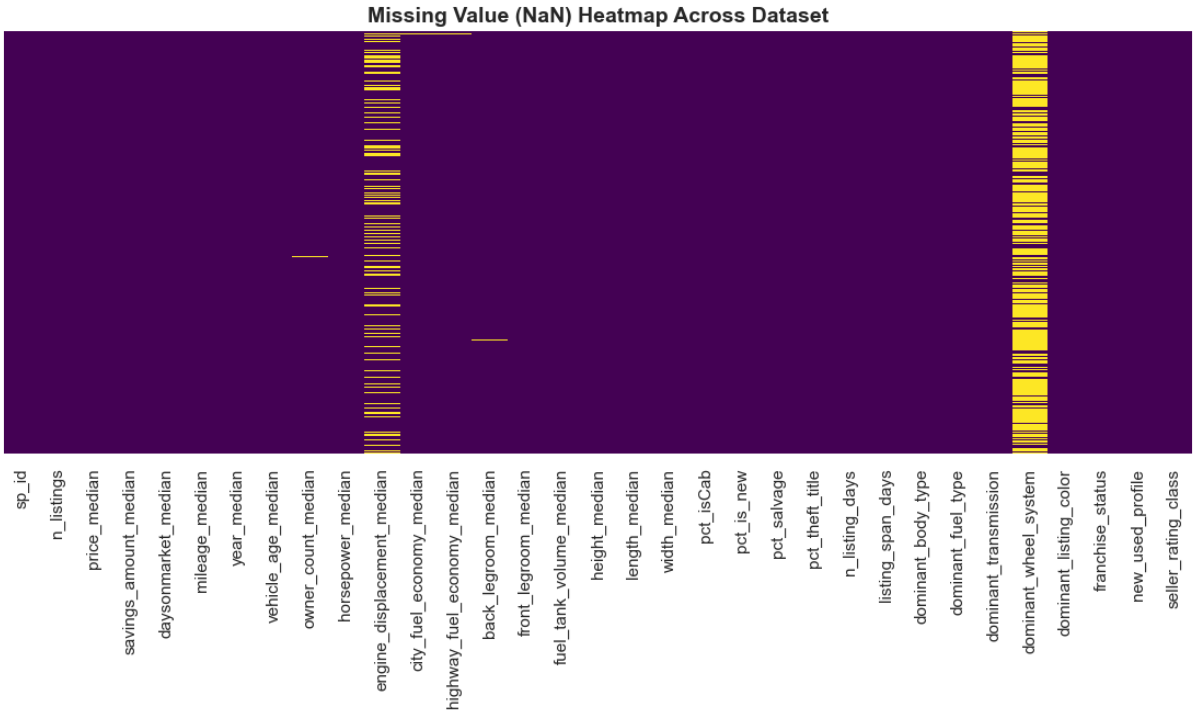


Figure 4: Missing-value (`NaN`) heatmap across the dataset.

- The remaining 14 numerical attributes (`owner_count_median`, `pct_isCab`, `pct_salvage`, `pct_theft_title`, physical dimensions, legrooms, fuel economy, `horsepower_median` and `mileage_median`): minor missing values ($< 0.30\%$ each) are imputed using the group median across each `dominant_body_type`, preserving expected characteristics across body types.

1.3.2 Outlier and noise detection

We used boxplots to spot the presence of possible outliers (Figure 5). Many attributes exhibit a significant amount of outliers, so we investigated each of them to distinguish true measurement noise from valid domain outliers.

True semantic noise. We identified 3 instances with `back_legroom_median` $= 0$ although their `dominant_body_type` is `suv_crossover` or `sedan`. Since this is physically impossible, we converted these values to the group median by body type, as described in Section 1.3.1.

Valid domain outliers. First, we examined `pct_salvage` and `pct_theft_title`, which record 90.83% and 84.56% zeros, respectively. This results in IQR $= 0$, so any non-zero value is flagged as an outlier. However, these values represent valid second-hand or salvage dealerships rather than data errors, so we retained all records. Second, we examined extreme values across all attributes:

- `n_listings` (up to 4,959) corresponds to a large-volume dealer.
- `price_median` (up to \$236,438), `horsepower_median` (up to 630 HP) and `vehicle_age_median` (up to 60 years) may correspond to luxury, sport or classic-car dealers.
- `city_fuel_economy_median` (up to 126 MPG) represents electric-vehicle dealers.
- `daysonmarket_median` (up to 1,987 days) reflects real market heterogeneity, as some dealers have slow-moving inventory.

We decided to keep all records from these valid sellers to retain niche market segments. The cleaned dataset therefore contains 13,843 records. We then applied logarithmic tail compression in Section 1.3.3. For distance-sensitive models (such as K-Means), task-specific outlier filtering or robust scaling can be applied without altering the master dataset.

1.3.3 Variables transformation

To address the heavy right skewness across the numerical attributes, we applied a logarithmic transformation, $\log(1 + x)$, to the heavily right-skewed continuous features: `price_median`, `n_listings`, `daysonmarket_median`, `savings_amount_median`, `vehicle_age_median`, `city_fuel_economy_median`, `highway_fuel_economy_median`, `engine_displacement_median` and `fuel_tank_volume_median`. This compresses extreme right tails and restores near-symmetric distributions without undefined values at zero. We then performed feature scaling to ensure uniformity. We generated two standardised representations: Z-score standardisation to mitigate outlier leverage in distance-based algorithms, and Min-Max scaling for bounded metrics.

1.3.4 Pairwise correlations and variable elimination

We analysed inter-feature dependencies across the numerical attributes using the pairwise scatterplot matrix (Figure 6) and the Spearman correlation matrix (Figure 7). We selected Spearman over Pearson because Figure 6 reveals strong non-linear monotonic patterns; in particular, `price_median` drops sharply as `mileage_median` and `vehicle_age_median` increase and then flattens out, forming an L-shaped curve alongside the heavy right-tail outliers.

Careful observation of the correlation matrix reveals several important findings. First, regarding highly correlated pairs, we retained both `mileage_median` and `vehicle_age_median` (0.93) as well as `n_listings` and `n_listing_days` (0.96) in the master dataset, because their relative ratios capture annual vehicle usage and active posting intensity. Second, within the technical specifications, `engine_displacement_median` correlates positively with `fuel_tank_volume_median`

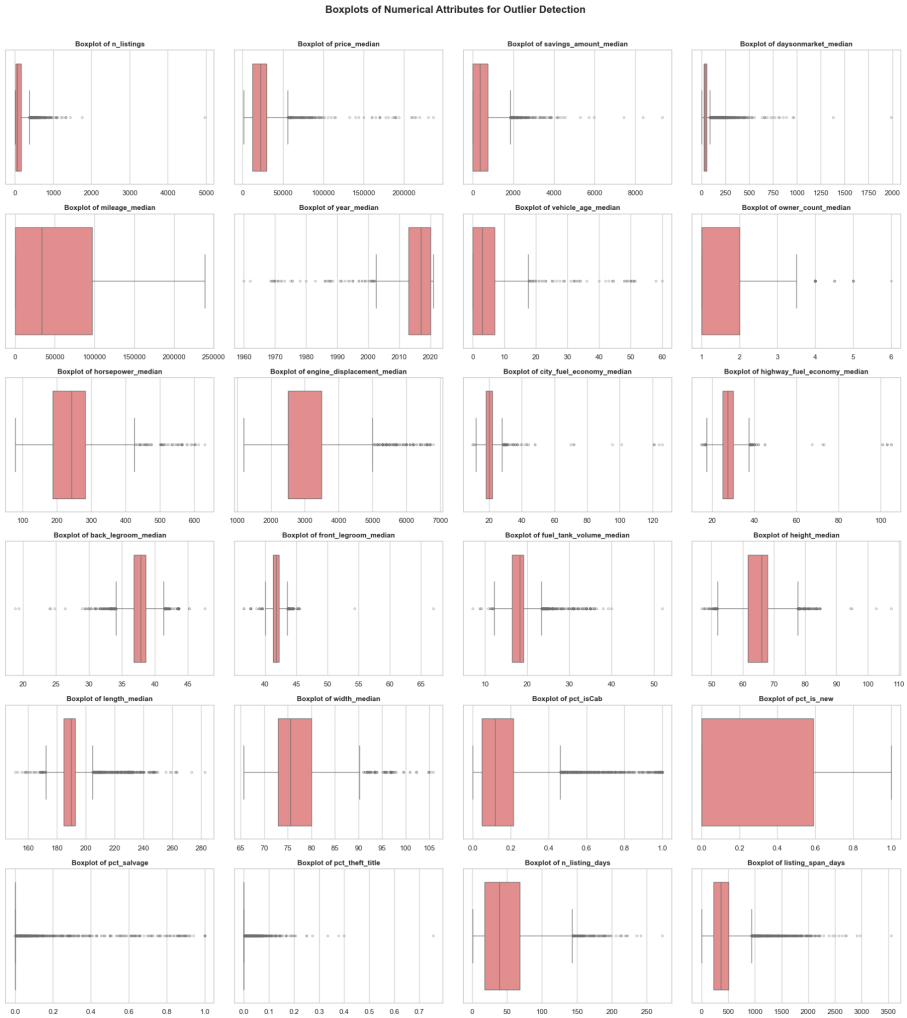


Figure 5: Boxplots of the numerical attributes for outlier detection.

(0.79) and `horsepower_median` (0.77), but negatively with `city_fuel_economy_median` (−0.76), reflecting the trade-off between engine capacity and fuel efficiency.

Finally, we removed `year_median` (−0.98 with `vehicle_age_median`, since the latter equals 2020−`year_median`) and `highway_fuel_economy_median` (0.94 with `city_fuel_economy_median`, as they describe the same fuel efficiency). The final prepared dataset therefore contains 13,843 rows and 29 attributes.

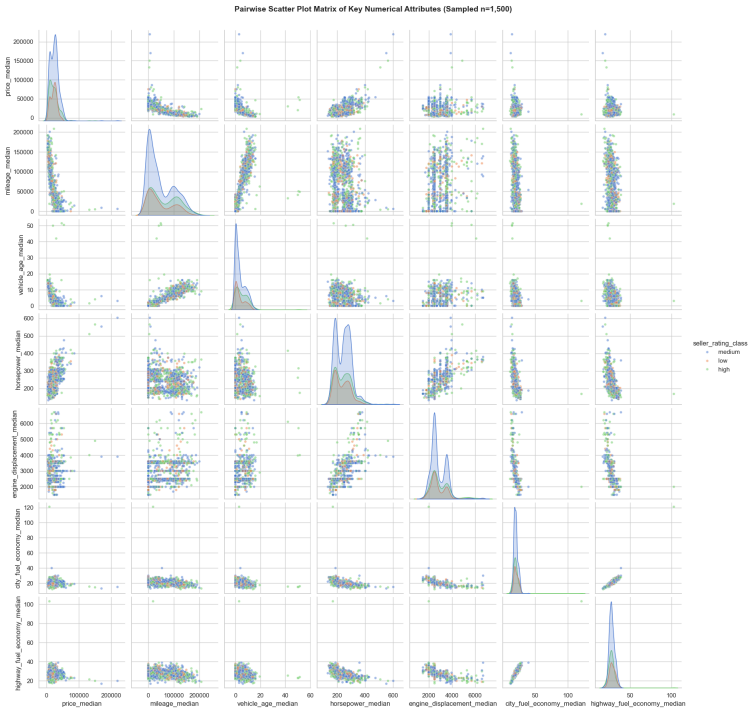


Figure 6: Pairwise scatter-plot matrix of the key numerical attributes (sampled).

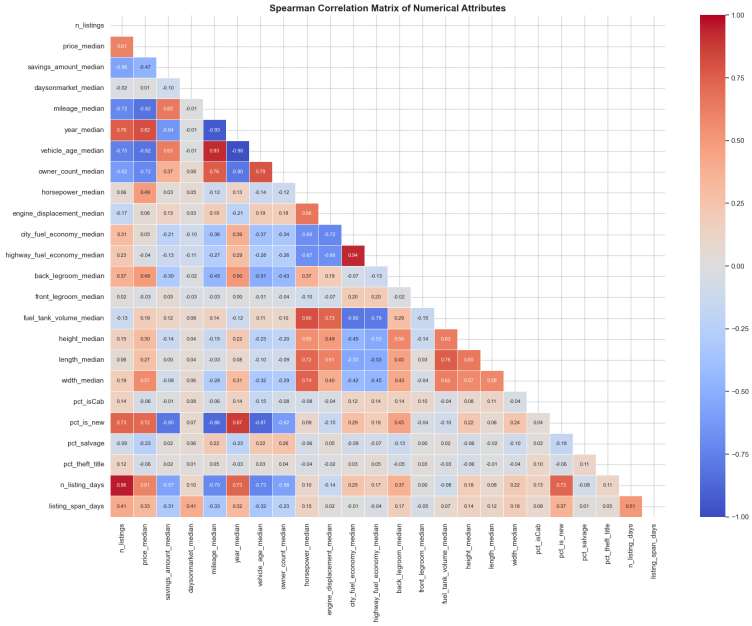


Figure 7: Spearman correlation matrix of the numerical attributes.

2 Clustering

(TODO: describe the clustering setup, the features used, and how dimensionality reduction was applied.)

2.1 Centroid-based

(TODO: K-Means; elbow and silhouette for choosing K.)

2.2 Density-based

(TODO: DBSCAN; k-NN distance plot to choose ε and MinPts.)

2.3 Hierarchical

(TODO: agglomerative clustering; linkage comparison and dendrograms.)

2.4 Final discussion

(TODO: compare the methods and state which worked best.)

3 Classification

The target variable is `seller_rating_class`. Describe the preprocessing, the train/test split, and the evaluation metrics (accuracy, precision, recall, F1, ROC-AUC).

3.1 KNN

(TODO: hyperparameter search results and per-class metrics.)

3.2 Naïve Bayes

(TODO: Gaussian and/or Categorical Naive Bayes results.)

3.3 Decision Trees

(TODO: tuning, pruning, and the resulting tree.)

3.4 Final conclusion

(TODO: compare all classifiers and recommend one.)

Table 4: Model comparison. *(Fill in.)*

Model	Accuracy	Macro-F1	Macro-AUC
k-NN			
Naïve Bayes			
Decision Tree			

4 Regression

(TODO: describe the regression target and predictors.)

4.1 Univariate regression

(TODO: simple regression per predictor.)

4.2 Multivariate regression

(TODO: multiple and multi-target regression.)

5 Pattern Mining

(TODO: describe the discretization/binning of numeric attributes and the transactional representation.)

5.1 Frequent patterns

(TODO: Apriori vs FP-Growth, support and itemset-size analysis.)

5.2 Association rules

(TODO: top rules, support/confidence/lift, and interpretation.)

6 Conclusion

(TODO: summarise the findings and the main takeaways.)

Team Contributions

Member	Contribution
Muhammad Iqbal Zain	
Truong Giang Do	
Vladislav Zhelezov	